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AC CIRCUIT THEORY

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1. INTRODUCTION

Alternating Current (AC) circuit theory focuses on the mathematical analysis of electrical circuits in which the voltages and currents applied to them varies periodically with time. The variation of an alternating voltage or an alternating current over time is called a waveform. That is an AC Waveform.

An alternating current waveform can take on many shapes and forms, but the one we are most interested in here, and which delivers electrical power to our homes is the sinusoidal waveform. Sinusoidal waveforms are more generally known as Sine Waves.

Direct Current (DC) voltage or current supplies can be characterised by a fixed magnitude (amplitude) and a definite direction with respect to time, making it a “Uni-directional” waveform. For example, +12V represents 12 volts flowing in a positive direction, while -5V represents 5 volts flowing in a negative or opposite direction around a circuit.

An alternating current waveform on the other hand is defined as one that varies in both magnitude and direction in more or less even manner with respect to time.

Sinusoidal waveforms are produced by rotating a coil of wire at a constant angular velocity within a fixed magnetic field

Thus an alternating waveform is a periodic waveform which alternates between some positive and negative value (usually many times per second) making it a “Bi-directional” waveform.

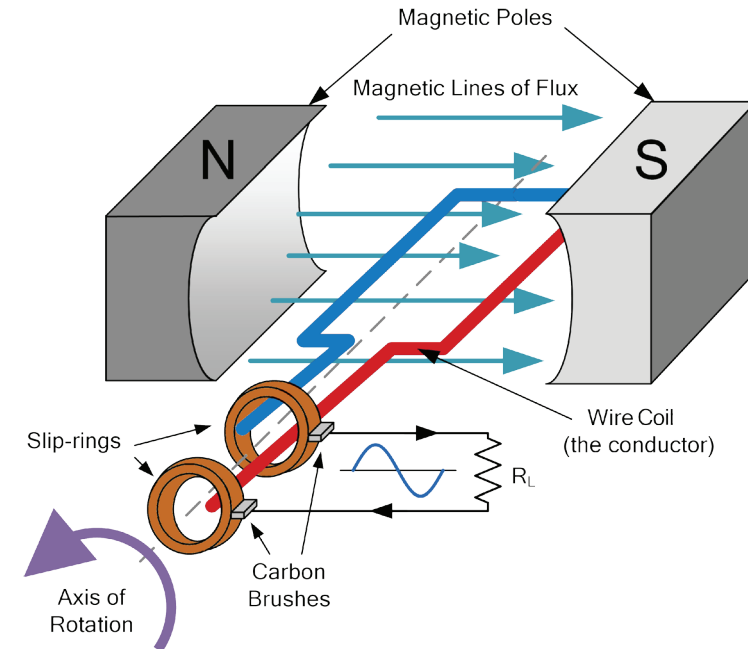
Alternating current waveforms are commonly produced by generators and by convention, alternating currents are called AC currents and alternating voltages are called AC voltages. The most common AC source is the commercial AC power that supplies energy to our homes and workplaces.

The AC Waveform is a “time-dependent signal” with the most common type of time-dependant signal being that of the Periodic Waveform. The periodic or AC waveform is the resulting product of a rotating electrical generator.

2. AC WAVEFORM GENERATION

Consider a basic AC generator consisting of a single loop of wire rotating within a magnetic field produced by two permanent magnets as shown in Figure 1.

FIGURE 1. A SIMPLE AC GENERATOR

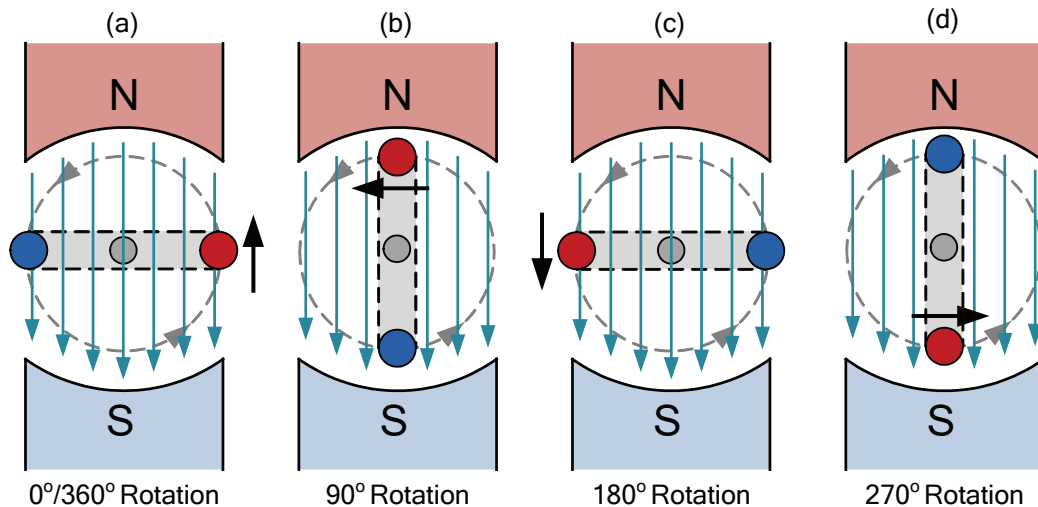


The basic AC generator shown in Figure 1, consists of a single loop or coil of wire which is allowed to rotate within a magnetic field produced by the poles of two magnets. For the ease of explanation, the coil has been divided into two halves. One half red, the other half blue rotating anti-clockwise around a central axis as shown.

The two ends of the coil are connected to slip rings which also rotate with the coil, with the electrical contact between coil and external circuit is made by two carbon brushes touching these slip rings.

When the coil rotates (moves) around its fixed axis, it cuts through the magnetic lines of flux present between the north and south poles of the magnets, firstly in one direction and then through the other direction as demonstrated below in Figure 2.

FIGURE 2. ROTATION OF A COIL WITHIN A MAGNETIC FIELD



The coil of conductive wire moves through the permanent magnetic field cutting through the lines of flux as it rotates, and as such an emf (electro-motive force) is induced within it. Starting with Figure 2a, the conductor of the coil is moving upwards and “in parallel” with the magnetic field. Therefore, at 0° rotation (start) no lines of flux are being cut and therefore no emf (voltage) is being induced into the conductor wire.

As the conductor continues to rotate towards the 90° position of Figure 2b, the conductor now moves at right angles (perpendicular) to the magnetic field. As it cuts perpendicular across the lines of flux, the maximum amount of magnetic flux is being cut at this point thus producing the maximum amount of induced emf in the coil.

Thus as the conductor rotates through the magnetic field at different angles between 0° and 90° the amount of induced emf in the coils conductor will lie somewhere between zero at 0° and its maximum emf value at 90° .

The conductor now continues to rotate towards the position of Figure 2c. As it does it again cuts through fewer and fewer lines of magnetic flux. This means that the induced emf in the coils conductor begins to decrease until at position 2c. Position 2c is 180° from the coils original start position, so the conductor is again moving in parallel with the magnetic field. So as before, since there are no lines of magnetic flux being cut, there is no emf voltage being induced into the conductor. The same situation as for Figure 2a.

The coil continues to rotate through the other half revolution between 180° and 360° (360° being one full revolution). Thus an equal amount of emf is induced within the coils conductor as previously seen except that this time the polarity is reversed.

An Alternating Current (AC) waveform is defined as one which changes periodically both in magnitude and direction

Thus as the conductive coil rotates through one half a revolution the polarity is positive in value, and as it rotates through the other half revolution the polarity is negative in value.

Then the amount of EMF induced within a conductor depends on the angle between the conductor and the magnetic flux as well as the strength of the magnetic field. Also, the rotation of the coils conductor through a full 360° of rotation within a magnetic field results in the periodic reversal of its polarity and the creation of a steady-state AC sinusoidal waveform representing one complete cycle (0° to 360°).

3. SINUSOIDAL FREQUENCY

As one cycle of induced EMF is produced in the conductor each full revolution (0 to 360°) of the coil through the magnetic field, if the coil rotates at a constant speed a constant number of cycles will be produced every second. The time taken for the coil to complete one full cycle is known as the waveforms periodic time, (T) in seconds.

The number of times one complete cycle (revolution) appears in one second of time is commonly known as the frequency, f (measured in Hertz). So 50 hertz means 50 cycles per second, and 100 hertz means 100 cycles per second. The relationship between periodic time and frequency is given as:

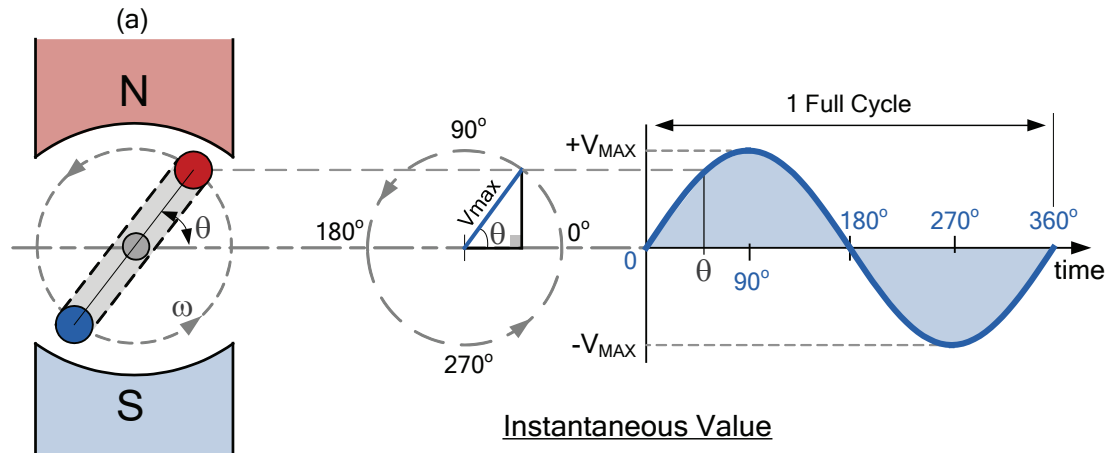
$$\text{Frequency, } f = \frac{1}{\text{Periodic Time}} = \frac{1}{T} \text{ hertz (Hz)}$$

So we can say that frequency is the reciprocal of the periodic time, as $f = 1/T$, or periodic time is the reciprocal of frequency as $T = 1/f$.

4. INSTANTANEOUS OR EFFECTIVE VALUE

The output voltage from our simple AC generator can be expressed over time in one of two ways. Graphically as a sinusoidal waveform, or algebraically by the use of a mathematical equation.

FIGURE 3. GRAPHICAL REPRESENTATION OF A SINE WAVE



Rotation starts at zero, 0° increases to a positive maximum (peak) value at 90°, and then decreases back down to zero at 180°. Here it changes polarity, increases to a negative maximum value at 270°, then returns again back to zero at 360°. One complete variation between the same points on the waveform is one cycle and since the waveform repeats itself at regular intervals over time. This is therefore known as a periodic waveform.

The instantaneous value of an AC voltage or current waveform, is its value at any one particular “instant” in time. The value at that instant maybe zero or maximum, either positive or negative. The instantaneous value of the sine wave is generally given as the “Instantaneous value = Maximum value x sin θ ”. Thus it is a time variable $a_{(t)}$, so there can be an infinite number of instantaneous values between 0° and 360°. One full revolution.

The instantaneous value of a voltage waveform which varies sinusoidally with time can be represented as:

$$v_{(t)} = V_{MAX} \sin \theta$$

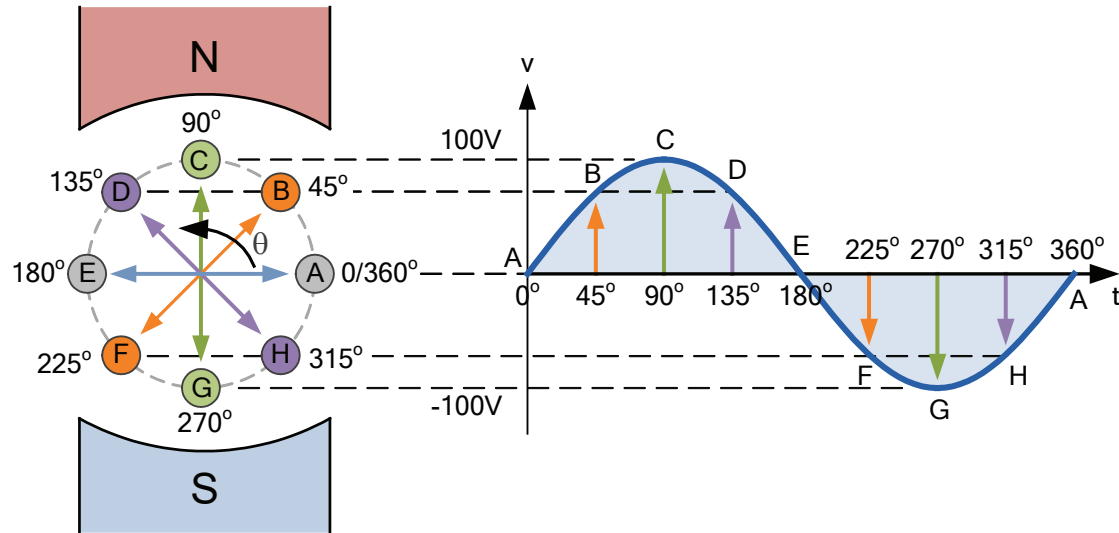
Where $v_{(t)}$ is the instantaneous voltage at time, (t). V_{MAX} is the maximum or peak value of the waveform, and θ is the angle of rotation. Since sinusoidal waveforms vary with time, AC supplies are commonly designated by lowercase letters $v(t)$ for voltage, and $i(t)$ for current instead of uppercase letters V and I for steady state DC values. Note that the subscript (t) here represents time.

If we know the maximum or peak value of the waveform, by using the formula above the instantaneous values at various points along the waveform can be calculated. By plotting these values out onto graph paper, a sinusoidal waveform shape will be constructed.

In order to keep things simple, we will plot the instantaneous values for the sinusoidal waveform at every 45° of rotation giving us 8 points to plot. Again, to keep it simple we will assume a maximum voltage, V_{MAX} value of 100V. Plotting the instantaneous values at shorter intervals, for example at every 30° (12 points) or 10° (36 points) for example would result in a more accurate sinusoidal waveform construction.

FIGURE 4. ALGEBRAIC REPRESENTATION OF A SINE WAVE

Coil Angle	0	45	90	135	180	225	270	315	360
$v_{(t)} = v_{MAX} \sin \theta$	0	70.71	100	70.71	0	-70.71	-100	-70.71	0



The points on the sinusoidal waveform are obtained by projecting across from the various positions of rotation between 0 and 360° to the ordinate of the waveform that corresponds to the angle, θ . Therefore, a sinusoidal waveform has a positive peak at 90° (position C) and a negative peak at 270° (position G). Positions B, D, F and H, the EMF value corresponds to the formula: $v_{(t)} = V_{MAX} \sin \theta$.

Suppose we want to know the instantaneous voltage of a 60 volt peak (maximum) sinusoidal voltage at an angle of 30 degrees into the cycle.

$$v_{(t)} = V_{MAX} \sin(\theta)$$

$$v_{(t)} = 60 \sin(60^\circ)$$

$$v_{(t)} = 60 \times (0.866)$$

$$v_{(t)} = 52 \text{ volts}$$

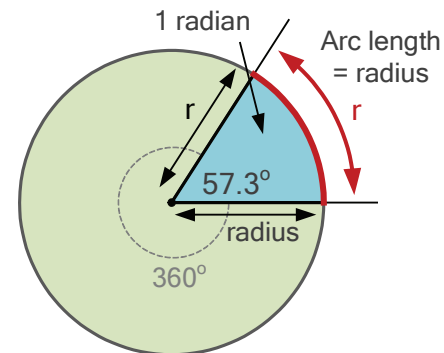
So the instantaneous voltage of the sinusoidal waveform at an angle of 30 degrees is 52V.

5. RADIAN UNITS OF A SINUSOIDAL WAVEFORM

The coil of the previous simple generator rotates around a central axis and completes one full cycle every revolution and up to now we have said that this full revolution or cycle is equal to 360°. However, there are other ways of measuring an angle of a circle other than degrees.

But when dealing with sinusoidal waveforms in the time domain and especially current related sine waves the unit of measurement used along the horizontal axis of the waveform can be either time, degrees or radians. In electrical engineering it is more common to use the Radian as the angular measurement of the angle along the horizontal axis rather than degrees.

FIGURE 5. DEFINITION OF A RADIAN



The radian (rad), is defined mathematically as the quadrant of a circle where the distance subtended on the circumference of the circle is equal in length to the radius (r) of the circle as shown in Figure 5.

The arc created along the circumference is exactly equal in length to the radius (r) of the circle. Thus, the angle θ is equal to one radian.

The length of one radian (r) will fit $6.284 (2 \times \pi)$ times around the whole circumference of the circle, so there must be 2π radians around the 360° of a circle. In other words, **360° = 2π radians**. Therefore one full radian must be equal to $360^\circ / 2\pi$ which equals to an angle of 57.3°.

As one full cycle of a sinusoidal waveform (sine wave) is created when a coil rotates a full 2π radians, the time (t) taken for the coil to rotate through an angle (θ) of the circle is therefore given as θ/t . This represents the coils angular velocity (ω) which is commonly given in radians/second, or rads/sec.

Knowing that the constant of pi (π) is equal to 3.142, (the c/d ratio of circumference, c to diameter, d), the relationship between the number of radians in a 360° angle to the number of degrees for a sinusoidal waveform is therefore given as:

5.1 Conversion Between Degrees and Radians

$$\text{Radians} = \left(\frac{\pi}{180} \right) \times \text{degrees}$$

$$\text{Degrees} = \left(\frac{180}{\pi} \right) \times \text{radians}$$

Degrees	0°	45°	90°	135°	180°	225°	270°	315°	360°
Radians	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	$\frac{3\pi}{4}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π

So if one full cycle of the sine wave is created when the coil rotates 2π radians, the mathematical relationship between periodic time, T , angular velocity, ω and frequency, f of a sinusoidal supply in an AC circuit is therefore:

$$\omega = 2\pi f = \frac{2\pi f}{T} \text{ rads/sec}$$

Thus, the angular frequency of a 50Hz supply will be: 314.2 radians/sec, and for a 60Hz supply: 376.8 radians/sec.

We can now take the instantaneous value of the previous voltage waveform which varies sinusoidally with time and express it in terms of angular velocity as being:

$$v_{(t)} = V_{\text{MAX}} \sin(\omega t)$$

Note that the Instantaneous Current that flows around a circuit as a result of the applied sinusoidal voltage will also be a sinusoidal waveform. The maximum (peak) value of current in an AC circuit is simply the peak value of voltage divided by the impedance (resistance) in the circuit.

Therefore, the instantaneous value of the current at any given electrical degree point along the current waveform can be calculated in exactly the same way as for the instantaneous voltage.

6. PHASE DIFFERENCE BETWEEN TWO SINE WAVES

Previously we saw that a sinusoidal waveform is an alternating quantity that can be presented graphically in the time domain along a horizontal axis. We also saw that as an alternating quantity, sine waves have a positive maximum value at time $\pi/2$, a negative maximum value at time $3\pi/2$, with zero values occurring along the baseline at 0, π and 2π .

However, not all sinusoidal waveforms will pass exactly through the zero axis point at the same time, but may be “shifted” to the right or to the left of 0° by some value in degrees when compared to another sine wave of the same frequency. For example, comparing a voltage waveform to that of a current waveform.

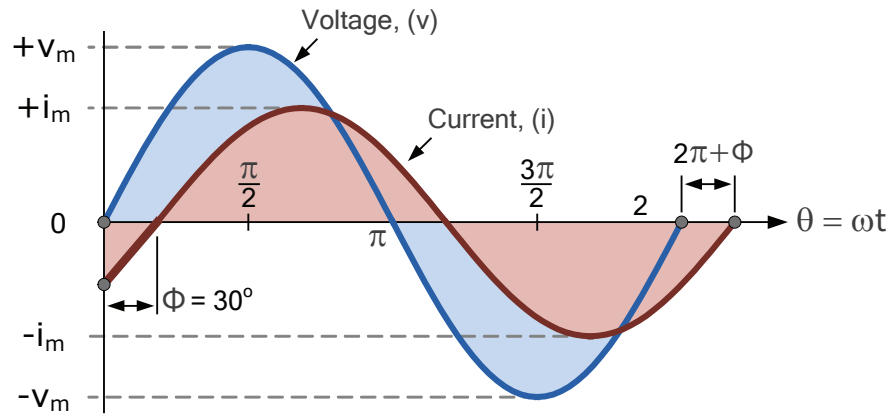
This then produces an angular difference or Phase Difference between the two sinusoidal waveforms.

Any sine wave that does not pass through zero at $t = 0$ when referenced to another has a phase shift. Then we can correctly say that one waveform “leads” or “lags” the other waveform by some phase angle of Φ .

Note that only a phase difference of less than 180° specifies whether a waveform leads or lags another waveform and are therefore “out-of-phase” with each other. Zero-degree phase difference indicates that the sine waves are “in-phase” with each other. Consider Figure 6.

Phase shift is the angular difference between the crossover points of two sine waves of the same frequency

FIGURE 6. PHASE DIFFERENCE BETWEEN SINUSOIDAL WAVEFORMS



The voltage, (v) and the current, (i) have a phase difference between them of 30° , so ($\Phi = 30^\circ$ or $\pi/6$ radians). As both alternating quantities rotate at the same angular velocity, i.e. they both rotate at the same frequency, this phase difference will remain constant for all instants in time. Generally, this phase difference of 30° between the two sine waves is represented by phi, Φ as shown.

Thus, if the voltage waveform equals: $v_{(t)} = V_{MAX} \sin(\omega t)$

Then the current waveform equals: $i_{(t)} = I_{MAX} \sin(\omega t - 30^\circ)$

Where the current, (i) "lags" (minus sign) the voltage, (v) by phase angle (Φ) of 30° .

Then we can see that phase angle (or phase shift) is just a convenient way of describing the amount of angular displacement between one sine wave to another wave of the same frequency along the same horizontal axis.

In the example of Figure 6. above, the angular displacement is: 30° or $\pi/6$ radians.

7. THE PHASOR DIAGRAM OF A SINE WAVES

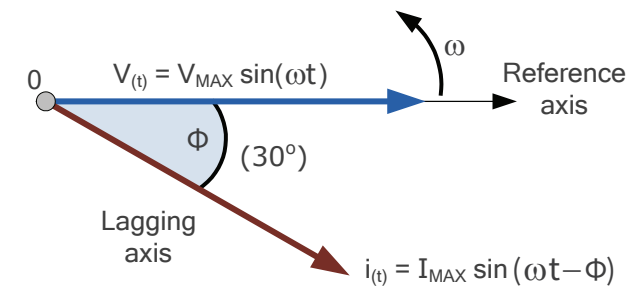
The generalized sinusoidal expression for an AC quantity is given as: $A_{(t)} = A_{MAX} \sin(\omega t \pm \Phi)$ representing the sinusoid in the time-domain form and where A (amplitude) represents a voltage or current waveform.

But when presented mathematically in this way, it is sometimes difficult to visualise this angular or phasor difference between two or more sinusoidal waveforms. One way to overcome this problem is to represent the sinusoids graphically within the spatial or phasor-domain form by using Phasor Diagrams, and this is achieved by the rotating vector method.

A rotating vector, simply called a "Phasor" is a scaled vector line whose length represents an AC quantity that has both magnitude ("peak amplitude") and direction ("phase") which has been "frozen" at some point in time. Then a phasor is a quantity that has both "Magnitude" and "Direction".

Consider the two waveforms from Figure 6 where the current, i is lagging the voltage, v by the angle Φ of 30° . The resulting phasor diagram will look like this.

FIGURE 7. PHASOR DIAGRAM FOR THE SINUSOIDAL WAVEFORMS



The phasor diagram of Figure 7. is normally drawn corresponding to a particular time along the horizontal axis. The lengths of the phasors are proportional to the rms values of the voltage, (v) and the current, (i) at the instant in time that the phasor diagram is drawn. The current phasor lags the voltage phasor by 30° in the direction shown.

The two phasors rotate together in an anticlockwise direction at the same angular frequency, then there is no relative motion between the two phasors. Therefore, the angle, Φ is also measured in the same anticlockwise direction. Thus a rotating vectors represent rms voltages, currents and their angular relationship to each other.

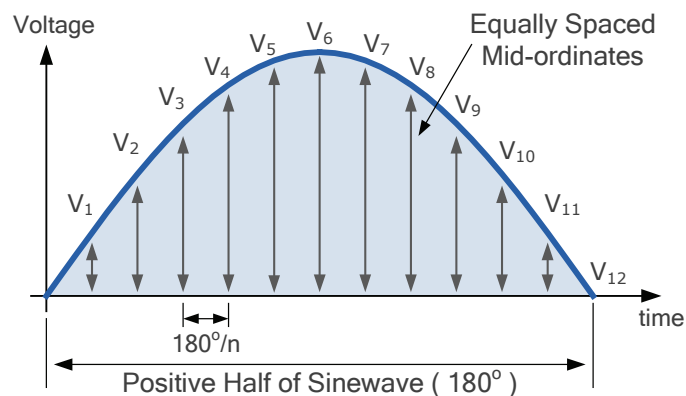
If the phase difference between the voltage and current is zero (0°), for example a purely resistive circuit, then the two vectors will be superimposed on top of each other. Note that Vectors, Phasors and Phasor Diagrams ONLY apply to sinusoidal AC alternating quantities.

8. THE AVERAGE OR MEAN VALUE OF A SINE WAVE

The average or mean value of a sinusoidal waveform is the average value measured over one half of a cycle only. This is because the average or mean value of a symmetrical sine wave will be zero, (0) since the area above the horizontal axis (the positive half cycle) is exactly the same as the area below the axis (the negative half cycle) and thus the two halves cancel each other out. Thus for a sinusoid, $A_{\text{AVERAGE}} = 0$.

In electrical terms, average or mean voltage, or even average current, can be used for both AC non-symmetrical waveforms or for DC rectification calculations. The symbols used for representing an average value are defined as being: V_{AV} or I_{AV} .

FIGURE 8. THE AVERAGE VOLTAGE GRAPHICAL METHOD



Consider only the positive half cycle between 0° and 180° (0 to π) from the previous sinusoidal voltage waveform. The average or mean voltage value of the waveform can be found graphically with a reasonable amount of accuracy simply by taking equally spaced instantaneous values along its length as shown in Figure 8.

The positive half of the waveform can be divided up into any number of "n" equally spaced portions, called "mid-ordinates". The width of each mid-ordinate will therefore be n° degrees (or "t" seconds) and the height of each mid-ordinate will be equal to the instantaneous value of the waveform at that point along the x-axis of the waveform as shown.

Each mid-ordinate value of the voltage waveform is added to the next and the summed total, V_1 to V_{12} is divided by the number of mid-ordinates used to give us the waveforms "Average Voltage" value. Then the average voltage (V_{AV}) is the mean sum of mid-ordinates of the voltage waveform and is calculated as:

$$V_{\text{AV}} = \frac{V_1 + V_2 + V_3 + V_4 + V_5 + \dots + V_{11} + V_{12}}{12}$$

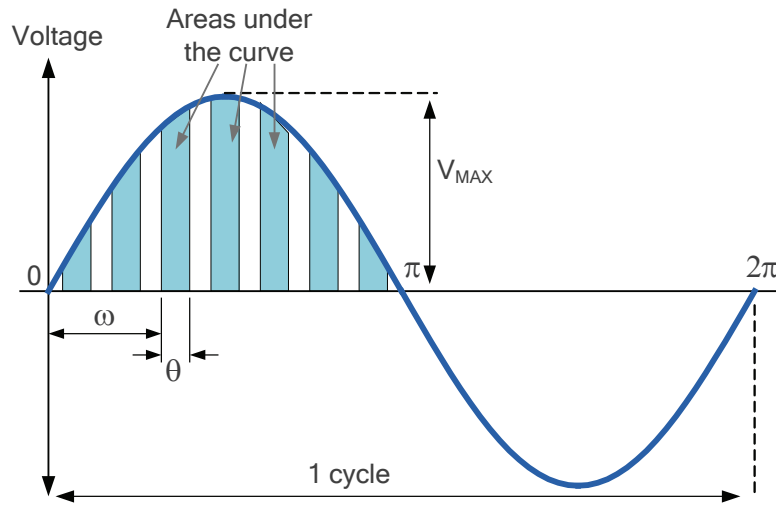
Here in this example, the average (mean) value of the symmetrical waveform is calculated using only twelve mid-ordinates. A more accurate value can be obtained by using more mid-ordinate points along the axis.

The average value can also be calculated mathematically by taking the approximation of the area under the curve at various intervals to the distance or length of the base using rectangles. The area under the curve can be found using various mathematical approximation rules such as the trapezoidal rule, the mid-ordinate rule or Simpson's rule.

By approximating the areas of the rectangles under the curve as shown in Figure 9. we can obtain a rough idea of the actual area of each one. So by adding together all the calculated areas the average value can also be found. If an infinite number of smaller thinner rectangles were used, the more accurate would be the final result as it approaches π .

The average value of a sinusoid is zero as the two half-cycles cancel out

FIGURE 9. THE ANALYTICAL APPROXIMATION OF AREA METHOD



The mathematical area under the positive half cycle of the periodic wave which is defined as $V_{(t)} = V_{MAX} \sin(\omega t)$ and having a period of T of one cycle using integration is given as:

$$V_{AV} = \frac{1}{\pi} \int_0^{\pi} V_{MAX} \sin \theta \, d\theta$$

$$V_{AV} = \frac{V_{MAX}}{\pi} (-\cos \theta)_0^{\pi}$$

$$V_{AV} = \frac{2V_{MAX}}{\pi} = \frac{2}{\pi} V_{MAX} = 0.637 V_{MAX}$$

Thus, the standard mathematical equation given for the average or mean voltage (or current) value of a symmetrical sinusoidal waveform is: $V_{AV} = 0.637 V_{MAX}$ (Note that V_{MAX} can also be represented as V_{PEAK} , or V_p). This $0.637 V_{MAX}$ average value of a sinusoidal waveform is also known as its equivalent d.c. value.

9. THE ROOT MEAN SQUARED OR RMS VALUE OF A SINE WAVE

Since neither the maximum peak value (V_{MAX}) nor the average value (V_{AVE}) can define the useful work done by a sine wave with regards to its I^2R heating effect compared to an equivalent DC power supply. The effective value which would cause the same amount of power to be dissipated over one complete cycle of the waveform is known commonly as the **Root-Mean-Squared**, or **RMS** value of the of the sine wave.

In other words, the rms value is an equivalent DC value which tells you how many volts or amps of DC that a time-varying sinusoidal waveform is equal too in terms of its ability to produce the same amount of electrical power and therefore work done.

We can define the term used to describe an rms voltage (V_{RMS}) as being “the square root of the mean of the square of the mid-ordinates of the voltage waveform” and this is given as:

$$V_{RMS} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^2 + V_4^2 + V_5^2 \dots + V_{11}^2 + V_{12}^2}{12}}$$

Whilst the method of calculation is the same for both halves of a sinusoidal waveform, for this rms example we will consider only the positive half cycle the same as for the previous average voltage example of Figure 8.

Using the “mid-ordinate rule” of the previous example, each mid-ordinate value of the voltage waveform (same applied for current waveform) is multiplied by itself (squared) and added to the next. This method gives us the Squared part of the rms voltage expression.

Next this squared value is divided by the number of mid-ordinates used to give us the Mean part of the rms voltage expression, and in our simple example above the number of mid-ordinates used was twelve (12). Finally, the square root of the previous result is found to give us the Root part of the rms voltage.

Again, the rms value can also be calculated mathematically by taking the approximation of the area under the curve of Figure 9. at various intervals.

$$V_{RMS} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} V_{MAX}^2(\omega t) dt}$$

$$V_{RMS} = \sqrt{\frac{V_{MAX}^2}{\pi} \int_0^{\pi} \frac{1 - \cos(2\omega t)}{2} d(\omega t)}$$

$$V_{RMS} = \frac{\sqrt{2V_{MAX}^2}}{2} = \frac{1}{\sqrt{2}} V_{MAX} = 0.7071 V_{MAX}$$

Therefore, the standard equation given for the effective or rms voltage (or current) value of a sinusoidal waveform is: **$V_{RMS} = 0.7071 V_{MAX}$** (again as stated previously, V_{MAX} can also be represented as V_{PEAK} , or V_p).

When dealing with AC voltages and currents, the rms value is the value that is always used. The domestic mains supply in the United Kingdom is 240 VAC, 50Hz. This value is assumed to indicate an effective value of “240 Volts rms”. Likewise, in the United States, the domestic supply is defined as being 120 Vrms, 60Hz.

10. FORM FACTOR AND CREST FACTOR

Although little used these days, both Form Factor and Crest Factor can be used to give information about the actual shape of an AC waveform. Form Factor (FF) is the ratio between the rms value and the average value of the alternating quantity.

$$\text{Form Factor (FF)} = \frac{\text{RMS Value}}{\text{Average Value}} = \frac{0.7071 V_{MAX}}{0.637 V_{MAX}}$$

For a pure sinusoidal waveform, the Form Factor (FF) will always be equal to **1.111**.

Crest Factor (CF) is the ratio between the waveform's peak value and it's rms value.

$$\text{Crest Factor (CF)} = \frac{\text{Peak Value}}{\text{RMS Value}} = \frac{V_{MAX}}{0.7071 V_{MAX}}$$

For a pure sinusoidal waveform, the Crest Factor (CF) will always be equal to **1.414**.

11. SINUSOIDAL WAVEFORM CONVERSION

For a sinusoidal waveform, the peak value V_p , it's average value V_{AVE} and the rms value V_{RMS} are all dependent on each other. But conversion between these various sinusoidal values can sometimes be confusing so the following table gives a convenient way of converting one sine wave value to another.

Convert From:	Peak	Peak-to-Peak	Peak	Peak	Average	Average	RMS	RMS
Multiply By:	2	0.5	0.7071	0.637	1.570	1.111	1.414	0.901
To Get Value:	Peak-to-Peak	Peak	RMS	Average	Peak	RMS	Peak	Average

12. COMPLEX NUMBERS REPRESENTATION IN RECTANGULAR FORM

Vectors are commonly used in AC circuit analysis to represent and show the angular relationship between voltage and current values in the form of a phasor diagram. Vectors which originate from a fixed point on an axes and rotate in an angular direction (usually anticlockwise) around this point are called phasors.

Single-frequency voltages and currents of AC circuits are usually written as phasors. Thus a phasor, as shown in Figure 7, is a way to represent the amplitude and phase angle of a sinusoidal waveform.

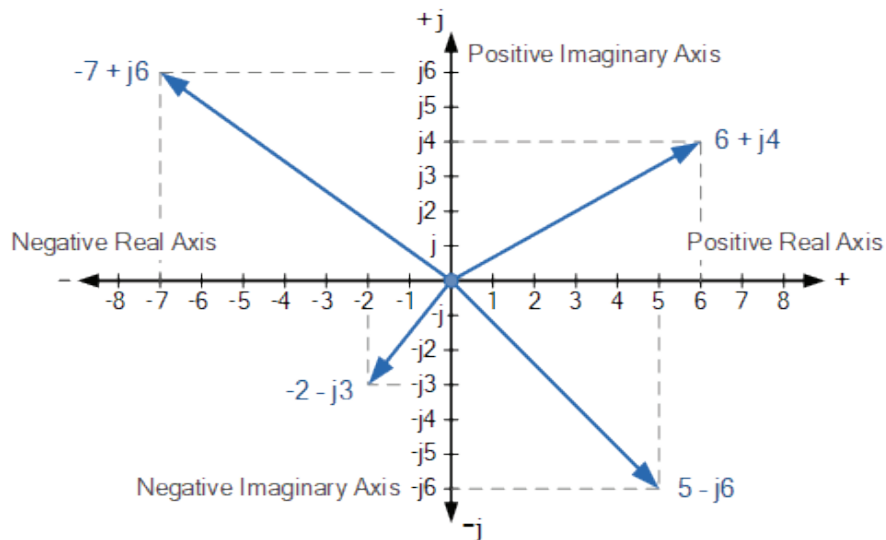
Unlike vectors, phasors can be represented by complex numbers, with the letter “j” representing being an imaginary number. This the letter “j” is placed in front of a real number to signify its imaginary number operation.

Examples of imaginary numbers are: j3, j12, j100 etc. Then a complex number consists of two distinct but very much related parts, a “Real Number” plus an “Imaginary Number”, and is presented in the form of: **$Z = a \pm jb$** .

Complex numbers in magnitude and phase form represent points in a two dimensional complex plane that are referenced along two distinct axes. A horizontal axis, called the “real axis”, and a vertical axis called the “imaginary axis”. The real and imaginary parts of a complex number can also be abbreviated to **$\text{Re}(z)$** and **$\text{Im}(z)$** , respectively.

Thus a complex number can be represented as a point on a two dimensional plane. So for example, $Z = 6 + j4$ represents a single point whose coordinates represent 6 along the horizontal real axis and 4 on the vertical imaginary axis. This is shown in Figure 10 along with other examples in all four quadrants.

FIGURE 10. RECTANGULAR FORM REPRESENTATION



Both the real and imaginary parts of a complex number in the “Rectangular Form” can be either a positive number or a negative number. Then both the real and imaginary axis must also extend in both the positive and negative directions producing a complex plane with four quadrants.

Note that complex numbers can also have “zero” real or imaginary parts such as: $Z = 6 + j0$ (pure 6Ω resistance) or $Z = 0 + j4$ (pure 4Ω reactance). In this case the points are plotted directly onto the real (horizontal) or imaginary (vertical) axis lines.

10.1 Addition and Subtraction of Complex Numbers

The addition or subtraction of complex numbers can be done either mathematically or graphically in rectangular form. For addition, the real parts are firstly added together to form the real part of the sum, and then the imaginary parts to form the imaginary part of the sum. For example:

$$Z_1 + Z_2 = (\pm \text{Re}_1 \pm \text{Re}_2) + j(\pm \text{Im}_1 \pm \text{Im}_2)$$

Assume we have two rotating vectors defined as, $Z_1 = 4 + j1$ and $Z_2 = 2 + j3$ respectively. The sum (addition) and the difference (subtraction) of the two vectors is given in both rectangular ($a + jb$) form and graphically as a phasor diagram.

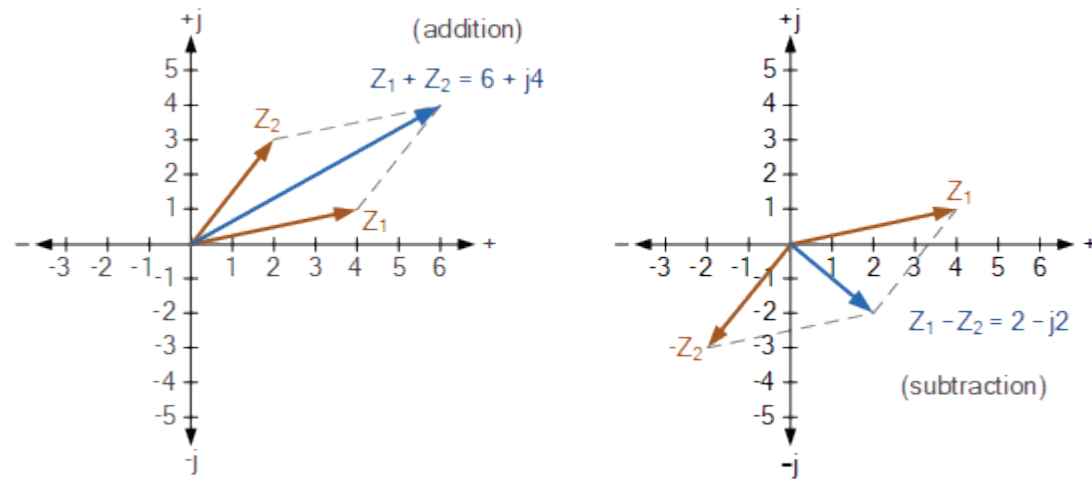
Addition of Two Phasors:

$$\begin{aligned} Z_1 + Z_2 &= (4 + j1) + (2 + j3) \\ Z_1 + Z_2 &= (4 + 2) + j(1 + 3) = 6 + j4 \end{aligned}$$

Subtraction of Two Phasors:

$$\begin{aligned} Z_1 - Z_2 &= (4 + j1) - (2 + j3) \\ Z_1 - Z_2 &= (4 - 2) + j(1 - 3) = 2 - j2 \end{aligned}$$

FIGURE 11. PHASOR REPRESENTATION



10.2 Multiplication and Division of Complex Numbers

The multiplication of complex numbers in the rectangular form follows more or less the same rules as for normal algebra along with some additional rules for the successive multiplication of the j -operator where: $j^2 = -1$. Then for the multiplication of the previous two vectors above: $Z_1 = 4 + j1$ and $Z_2 = 2 + j3$ the result is:

Multiplication of Two Phasors:

$$Z_1 \times Z_2 = (4 + j1) \times (2 + j3)$$

$$Z_1 \times Z_2 = 8 + j12 + j2 + j^2 3$$

As: $j^2 = -1$,

$$Z_1 \times Z_2 = 8 + j14 - 3 = 5 + j14$$

The division of complex numbers in rectangular form is a little more difficult to perform as it requires the use of the denominators conjugate function to convert the denominator of the equation into a real number.

The conjugate of a complex number is found by reversing the algebraic sign of the complex numbers imaginary part only while keeping the algebraic sign of the real part the same. For example, the complex conjugate of $z = 2 + j3$ is $z = 2 - j3$, likewise the conjugate of $z = 2 - j3$ is $z = 2 + j3$.

Division of Two Phasors:

$$\frac{Z_1}{Z_2} = \frac{4 + j1}{2 + j3} \times \frac{2 - j3}{2 - j3} = \frac{8 - j12 + j2 - j^2 3}{4 - j6 + j6 - j^2 9}$$

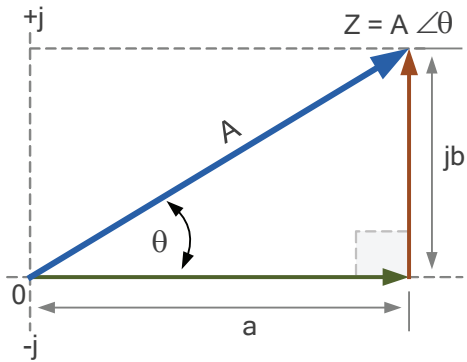
$$\frac{Z_1}{Z_2} = \frac{8 - j10 + 3}{4 + 9} = \frac{11 - j10}{13} = 0.85 - j0.77$$

13. COMPLEX NUMBERS REPRESENTATION IN POLAR FORM

The previous rectangular form plotted horizontal and vertical points in the complex plane. The *Polar Form* of a complex number is written in terms of its length (magnitude) and phase angle away from the central zero intersection.

Thus, a polar form vector $Z = a + jb$ is presented as: $Z = A \angle \pm\theta$, where: Z is the complex number in polar form, A is the magnitude or length of the vector and θ is the angle of A . Note that angle (θ) can be either positive or negative, but is always measured in an anticlockwise direction away from the positive real axis.

The magnitude and angle of the point still remains the same as for the rectangular form above, this time in polar form the location of the point is represented in a “triangular form” as shown in Figure 12.

FIGURE 12. POLAR FORM REPRESENTATION OF A COMPLEX NUMBER


$$Z = A \angle \pm \theta$$

$$A = \sqrt{a^2 + b^2}$$

$$\text{Also: } a = A \cos \theta \text{ and } b = A \sin \theta$$

$$\theta = \tan^{-1} \left(\frac{a}{b} \right)$$

The polar representation of a point is based around the triangular form. Then we can simply use geometry of the triangle and especially trigonometry and Pythagoras's Theorem on triangles to find both the magnitude and the angle of the complex number.

The basic mathematical operations of complex numbers in polar form are given as:

$$\text{Addition in Polar Form: } Z_1 + Z_2 = A_1 + A_2 \angle (\theta_1 + \theta_2)$$

$$\text{Subtraction in Polar Form: } Z_1 - Z_2 = A_1 - A_2 \angle (\theta_1 - \theta_2)$$

$$\text{Multiplication in Polar Form: } Z_1 \times Z_2 = A_1 \times A_2 \angle (\theta_1 + \theta_2)$$

$$\text{Division in Polar Form: } Z_1 \div Z_2 = A_1 \div A_2 \angle (\theta_1 - \theta_2)$$

$$\text{Complex conjugate in Polar Form is: } \sqrt{A_1} \angle \theta$$

14. CONVERTING BETWEEN COMPLEX NUMBER FORMS

Converting Polar to Rectangular Form ($P \rightarrow R$) of example: $6 \angle 30^\circ = a + jb$

$$a = A \cos \theta \quad \text{and} \quad b = A \sin \theta$$

$$6 \angle 30^\circ = (6 \cos 30^\circ) + j(6 \sin 30^\circ)$$

$$6 \angle 30^\circ = (6 \times 0.866) + j(6 \times 0.5)$$

$$\text{Therefore: } 6 \angle 30^\circ = 5.2 + j3$$

Converting Rectangular to Polar Form ($R \rightarrow P$) of example: $5.2 + j3 = A \angle \theta$

$$A = \sqrt{5.22 + 32} = \sqrt{36} = 6$$

$$\theta = \tan^{-1} (3 \div 5.2) = 30^\circ$$

$$\text{Therefore: } 5.2 + j3 = 6 \angle 30^\circ$$

Then, *Complex Numbers* consist of two distinct parts, a real number plus an imaginary number. Where the imaginary number is distinguished from the real number by using the j-operator. The multiplication of the j-operator by itself ($j \times j$) produces j^2 which is exactly equal to -1 resulting in a vector rotation of 180° .

In Rectangular Form a complex number is represented by a point in space on the complex plane and defined as: $Z = a + jb$. While in Polar Form a complex number is represented by a line whose length is the magnitude and by its phase angle being defined as: $Z = A \angle \pm \theta$.

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